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Does Financial Stability Withstand Economic Policy Uncertainty? Insights From Linear and Nonlinear Patterns

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ABSTRACT

This study examines the impact of Economic Policy Uncertainty (EPU) on Financial Stability (FS) in developed and emerging economies from 2003 to 2022. Using a balanced panel dataset and multiple statistical methods, the results reveal a nonlinear association between EPU and FS. At low to moderate levels, EPU reduces financial stability, but beyond a certain point, higher EPU improves FS. The findings suggest that rising EPU prompts regulators and institutions to respond in order to protect FS. Subsample analysis shows that this nonlinear effect becomes more pronounced in emerging economies, where financial systems remain more vulnerable and sensitive to shocks. Additional tests reveal that emerging markets respond more sharply to changes in uncertainty, while developed markets exhibit a weaker response, reflecting the strength of their financial systems. The study findings remain robust across a range of alternative measures and advanced estimators, including the generalized method of moments, spline regression, and panel-corrected standard errors. The study offers crucial policy insights by highlighting the need for tailored regulatory frameworks to mitigate the harmful effects of EPU on FS. By distinguishing between developed and emerging economies, this research enhances our understanding of financial resilience in the face of economic uncertainty and contributes to ongoing debates on macroeconomic stability.

JEL Classification: E60, G01, O16, O57, C32, C33

1 | Introduction

The stability of financial systems plays a vital role in sustaining economic growth, investor confidence, and overall prosperity. Over the past two decades, global financial markets experienced multiple shocks from the Global Financial Crisis (2007–2008) to the Covid-19 pandemic that exposed vulnerabilities in financial systems (Arif et al. 2020; Barik and Pradhan 2021; Leippold and Matthys 2022; Phan et al. 2021). These events underscored the significance of understanding how macroeconomic risks, particularly economic policy uncertainty (EPU), impact the resilience of financial systems across various economies. EPU refers to the unpredictability surrounding government policies, regulations, and decisions both domestically and internationally. Since Baker et al. (2016) introduced the EPU index, researchers have linked higher EPU to reduced investment, credit constraints, heightened volatility, and slowed economic growth (An et al. 2022; Julio and Yook 2016). However, despite its critical role, the impact of EPU on financial stability (FS)—the capacity of financial institutions and markets to withstand shocks—remains inadequately explored, especially in terms of complex, nonlinear dynamics.

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Financial stability refers to the ability of the financial system, which encompasses markets, intermediaries, and infrastructure, to withstand shocks that may disrupt the delivery of credit and payments to the real economy (BIS 2018; IMF 2019). This resilience may be compromised by periods of high Economic Policy Uncertainty (EPU), which alters the behavior of financial institutions and investors. Increasing uncertainty terms in lending decrease the supply of credit and elevate precautionary liquidity holdings, which affect systemic soundness. Policy clarity, on the other hand, brings about trust, investment, and asset stability. There is an ever-growing body of empirical evidence on this pathway of transmission between uncertainty and systemic fragility (Phan et al. 2021; Zhang and Wang 2023). Although the macroeconomic and market impact of EPU is increasingly discussed in the literature, the direct effects of EPU on financial stability remain under-researched. This nexus is important to understand in order to develop plausible macroprudential and fiscal systems in unstable settings. This paper aims to contribute to a more comprehensive understanding of EPU-induced financial fragility by empirically examining the effects of EPU and its nonlinear impacts on financial stability across both emerging and developed economies. This paper will complement the existing discussion of policy coordination in the new era of increased volatility.

The existing literature mainly adopts linear models to study the EPU–financial outcome nexus (Das and Ordal 2022; Phan et al. 2018), neglecting the possibility that the relationship might change at different levels of uncertainty. This is a critical oversight, as emerging theoretical and empirical evidence suggests a nonlinear or U-shaped relationship: while moderate uncertainty harms FS, very high uncertainty may prompt institutional reforms, regulatory tightening, and adaptive policy responses that restore stability (Aman et al. 2024; Ogbuabor et al. 2024; Wu et al. 2024). Another important but under examined dimension is cross-country heterogeneity. Developed and emerging economies differ substantially in institutional quality, financial market depth, and macroeconomic resilience (Liao et al. 2021; Orlowski 2023). Emerging markets often face greater EPU and weaker regulatory frameworks, making their financial systems more vulnerable to shocks compared to those in developed economies. Despite this, few studies explicitly compare how EPU influences FS differently across these economic groups.

This study argues that the EPU and FS relationship follows a nonlinear trajectory characterized by threshold effects, with more pronounced instability at certain levels of EPU but stabilizes beyond a critical point. This dynamic is expected to differ between developed and emerging markets, given variations in institutional capacity and macroeconomic and financial resilience. The theoretical foundation for this argument draws on Real Options Theory, which suggests that policy uncertainty delays investment and constrains liquidity, weakening FS in the short term (McDonald and Siegel 1986). Besides this, Financial Cycle Theory further explains how rising EPU amplifies downturns but may also trigger countercyclical interventions. Institutional Theory, as presented by Friedman (1970) emphasizes the role of central banks and governance in mitigating uncertainty through policy tools and regulatory oversight. Together, these frameworks suggest a U-shaped relationship, whereby FS initially deteriorates with rising EPU but may recover once institutional responses are activated.

Against this backdrop, this study investigates the nonlinear effects of EPU on FS across a cross-country setting. It examines how the EPU–FS relationship differs between developed and emerging economies, taking into account their institutional capacities, regulatory frameworks, and economic vulnerabilities. By integrating these cross-country distinctions, the study provides new insights into how varying levels of policy uncertainty shape FS and presents empirical evidence that captures the heterogeneity of financial systems under different macro-institutional environments.

The study examines a balanced panel of data from 12 countries: six developed (Australia, Canada, Denmark, Japan, the United Kingdom, and the United States) and six emerging (Brazil, China, India, Mexico, Russia, and South Korea), to reflect institutional, financial, and macroeconomic diversity. This selection addresses structural asymmetries. Developed economies possess mature financial systems, robust legal institutions, and credible policy frameworks that mitigate EPU shocks more effectively (Liao et al. 2021; Orlowski 2023). In contrast, emerging economies face greater exposure to policy volatility, weaker regulatory capacity, and higher macroeconomic fragility (Sarwar and Hassan 2021; Wang and Zhu 2025; Xu et al. 2021). Figure 1 presents the plot of FS in developed and emerging economies (monthly average), while Figure 2 shows the plot of EPU in the same groups. These differences provide a relevant context to assess how EPU interacts with FS across varying institutional settings. The selected countries also hold systemic importance in the global economy and represent a broad range of governance quality and policy regimes, which enhances the external validity of the findings. EPU is measured using the country-level index by (Baker et al. 2016), which reflects uncertainty in fiscal, monetary, regulatory, and trade policies. FS is proxied by two established indicators: the banking system *Z*-score (Sethy and Goyari 2022), which captures the probability of default and systemic strength, and the M2-to-GDP ratio (Broner et al. 2013; Miao and Qamruzzaman 2021; Yanikkaya 2003), used as a robustness measure to represent financial depth and liquidity.

In this paper, we apply a comprehensive set of econometric techniques, including ordinary least squares, fixed effects,

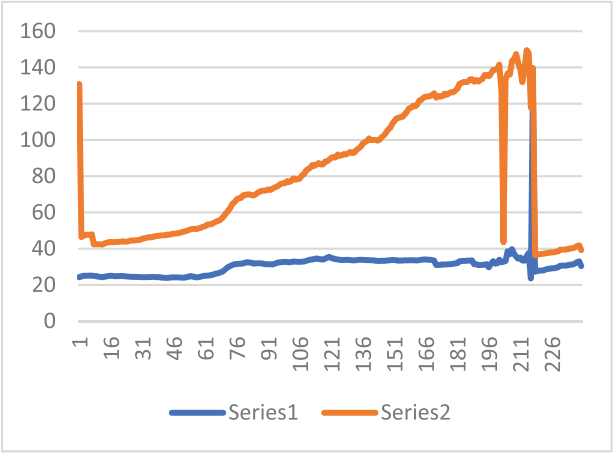


FIGURE 1 | Plot of FS in developed and emerging economies (monthly average).

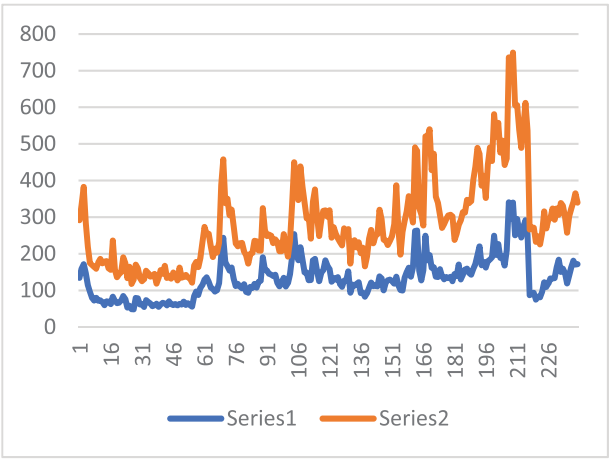


FIGURE 2 | Plot of EPU in developed and emerging economies (monthly average).

generalized least squares, quantile regression, spline regression, and dynamic panel generalized method of moments. Our analysis reveals a significant nonlinear association: EPU hurts FS at low to moderate levels but improves FS beyond a critical threshold. This nonlinear pattern appears markedly stronger in emerging economies, highlighting their heightened vulnerability and sensitivity to policy shocks compared to the more resilient financial systems of developed countries. These findings support and extend fundamental options theory and uncertainty-driven financial cycle models by demonstrating that uncertainty initially delays irreversible investments and destabilizes financial systems, but then triggers adaptive regulatory and institutional responses that restore stability.

Furthermore, this study estimates the long-run relationship using Fully Modified Ordinary Least Squares (FMOLS), which corrects for serial correlation and endogeneity. Results reveal a U-shaped relationship between EPU and FS, with emerging markets showing a lower threshold beyond which institutional adjustments enhance stability. Developed economies exhibit a similar pattern with a higher threshold, reflecting stronger institutional capacity. To address endogeneity and dynamic panel bias, we employ System GMM, which confirms the nonlinear relationship and validates the robustness of our results. Additionally, quantile regression highlights distributional heterogeneity: the negative impact of EPU on FS becomes more pronounced at higher stability levels, particularly in emerging economies. In contrast, developed economies show a positive effect at moderate FS levels.

The study makes several important contributions to the incumbent literature. First, it extends the literature by testing the nonlinear effects of EPU on FS and provides empirical evidence of threshold effects that earlier research often overlooks (Arif et al. 2020; Barik and Pradhan 2021; Leippold and Matthys 2022; Phan et al. 2021). For example, previous studies mainly assume a linear relationship between EPU and FS. However, this study reveals a U-shaped pattern where low levels of EPU reduce stability, while beyond a certain point, higher uncertainty leads to institutional reforms that strengthen FS. Desalegn et al. (2023) focus on the linear impact of EPU and bank competition on FS within China, limiting the analysis to a single country.

In contrast, this study examines the nonlinear effects of EPU on FS across multiple countries, thereby expanding the scope of the study. Second, it offers a comparative analysis between developed and emerging economies, showing that differences in institutional quality and regulatory frameworks create varying impacts of EPU on FS. Emerging markets experience greater instability under moderate uncertainty but benefit more from policy adjustments at high uncertainty levels, whereas developed economies demonstrate stronger resilience. Similarly, Ali et al. (2023) study a panel of 23 countries but do not test nonlinear relationships or distinguish between developed and emerging economies. This research fills that gap by explicitly modeling a U-shaped relationship and comparing these country groups based on their institutional differences. While Zhou et al. (2023) and Jia et al. (2021) examine the effects of EPU on firm-level innovation and bank stability, this study extends the focus to financial system stability at the macroeconomic level, offering a systemic perspective that captures cross-country heterogeneity. Third, the study applies advanced econometric methods such as Fully FMOLS, dynamic panel, and spline regression to improve estimation accuracy and address endogeneity concerns. Previous studies rely primarily on linear panel models or single estimation techniques. This study advances the literature by applying multiple robust econometric methods that enhance detection of threshold effects and handle endogeneity and heterogeneity. Together, these features enable the study to offer more nuanced insights and policy recommendations tailored to different institutional contexts and stages of economic development. Finally, the study offers practical policy insights by identifying threshold levels of EPU where its effect shifts from negative to positive. These findings help regulators design adaptive policies that enhance financial resilience under varying economic conditions.

The paper explicitly addresses the long-term impact of policy uncertainty on financial stability, unlike other researchers who have explored the subject using either the growth or market volatility channel to measure policy uncertainty. The research has three different contributions. First, it uses cross-sectionally augmented ARDL and Westernlund cointegration tests, which overcome a limitation of previous first-generation models associated with cross-sectional dependence. Second, it incorporates a nonlinear approach, as it includes EPU2, which enables the identification of threshold behavior in the relationship between uncertainty and stability. Third, it reveals the sensitivity of financial systems to policy shocks by comparing various economies through a single empirical framework. Combined, these contributions offer fresh insights into how uncertainty is transformed into systemic vulnerability.

The remainder of the article is organized as follows: Section 2 provides a concise literature review, Section 3 describes the data and methodology, Section 4 presents and discusses the results and robustness checks, and Section 5 concludes.

2 | Theoretical Framework and Hypothesis Formulation

This section outlines the theoretical foundations that support the correlation between EPU and FS. We elucidate the mechanisms

by which policy uncertainty affects FS based on established economic theories and empirical evidence.

2.1 | Theoretical Background

The relationship between EPU and FS is complex, drawing on theories from economics, finance, and political science. This framework provides the foundation for understanding how policy uncertainty influences financial systems, particularly in different economic contexts. Bernanke (1983) initially proposed that uncertainty negatively affects investment by increasing the option value of waiting. Firms tend to delay or reduce capital allocation under uncertain policy conditions. Baker et al. (2016) later operationalize this concept by introducing the EPU index, which captures ambiguity about policy decision-makers, timing, and expected economic consequences of actions or inaction. The Real Options Theory further explains that firms defer irreversible investment under uncertainty, leading to short-term instability in financial systems (McDonald and Siegel 1986). However, at higher uncertainty levels or over time, institutions may respond through regulatory reforms, monetary adjustments, or improved governance, which can gradually restore FS (Bae et al. 2025; Baker et al. 2023; Jurado et al. 2015). This creates a nonlinear relationship between EPU and FS, where the adverse effects may reverse after a certain threshold is reached.

Moreover, Monetary Policy Theory supports this framework (Friedman 1970). Central banks influence FS through interest rate changes and liquidity operations. Policy uncertainty raises volatility in credit markets and weakens investor sentiment, thereby undermining FS. Ko and Lee (2015) and Nguyen (2021) provide evidence that EPU affects stock markets and credit conditions, thereby intensifying systemic risk. These theoretical insights suggest a nonlinear relationship between EPU and FS. At the initial stages, higher uncertainty reduces investment, restricts credit availability, and increases financial risk. However, beyond a certain threshold often linked to institutional reforms or policy interventions FS may recover as markets reach a new equilibrium. This framework also accounts for structural differences between developed and emerging economies. Developed countries tend to maintain stronger institutional safeguards and regulatory systems, while emerging economies often face greater exposure due to weaker governance structures. As such, the EPU–FS nexus is likely to vary across country groups, reinforcing the need for a comparative empirical analysis.

2.2 | Empirical Literature

EPU has attracted considerable attention in both financial and economic literature due to its far-reaching effects on capital allocation, risk assessment, and market volatility. A vast body of research has explored the implications of EPU across macroeconomic and microeconomic domains. The evidence broadly suggests that higher EPU undermines investor confidence, slows down economic activity, and threatens financial stability. Numerous studies have documented the negative impact of EPU on capital flows (An et al. 2022; Julio and Yook 2016), corporate investment decisions (Agénor and Flamini 2022; Al-Thaqeb and Algharabali 2019; Julio and Yook 2016; Kang et al. 2014; Wang et al. 2014) and financial market dynamics (Ko and Lee 2015; Phan et al. 2018). These findings are often interpreted through the lens of fundamental options theory, which posits that heightened uncertainty encourages firms and investors to delay irreversible decisions in anticipation of more stable conditions.

On a broader scale, Balcilar et al. (2022) and Xu et al. (2021) show that increased EPU constrains economic growth. Similarly, Das and Ordal (2022) and Nguyen (2021) find that EPU weakens the resilience of the banking sector, particularly in major European markets. Using data from 23 countries between 1996 and 2016, Phan et al. (2018) find that higher EPU exacerbates financial instability. However, the study does not differentiate between developed and emerging economies, nor does it explore nonlinear dynamics. Stock markets also appear sensitive to EPU shocks. For example, Ko and Lee (2015) observe a substantial adverse effect of EPU on stock prices. Liao et al. (2021); Liu and Zhang (2015), and Ma et al. (2022) report that financial market volatility intensifies under higher uncertainty. Nguyen et al. (2020) show that EPU constrains credit growth, while Khan et al. (2020) conclude that policy uncertainty significantly affects capital structure decisions in US firms.

Although many of these studies examine linear relationships, a growing body of literature has begun to investigate nonlinear dynamics. For example, Nusair and Al-Khasawneh (2022) employed a nonlinear ARDL model to illustrate the asymmetric short- and long-run effects of EPU on the G7 stock markets. Tang et al. (2025), using MIDAS-CQR, show that EPU has more potent effects on financial stress during periods of elevated volatility. Khan et al. (2025) reveal that Japanese sectoral indices respond asymmetrically to EPU depending on market conditions and memory effects. These findings challenge the adequacy of linear models in capturing the full complexity of EPU's impact.

Other researchers focus on the interaction between EPU and institutional or firm-level conditions. For instance, Sarwar and Hassan (2021) find that board financial expertise moderates the effect of EPU on corporate dividend policy. Olasehinde-Williams and Olanipekun (2022) show that US EPU Granger-causes exchange market pressure in African economies, indicating cross-border vulnerabilities. Bagh et al. (2023) employ wavelet coherence analysis to examine the temporal dynamics between EPU and the Stock Price Index (SPI), utilizing phase difference methods to assess the alignment of their fluctuations. The findings reveal a strong and persistent correlation between EPU and SPI, particularly at low frequencies, highlighting a substantive connection between policy uncertainty and stock market behavior. Ogbuabor et al. (2024) document that rising and falling uncertainty asymmetrically affect real sector performance in Brazil, China, India, and South Africa. Fang et al. (2025) explore the mediating role of investor sentiment in the EPU–systemic risk relationship.

Despite significant advancements, critical gaps remain. Most studies focus mainly on the linear effects of EPU on financial outcomes (Das and Ordal 2022; Phan et al. 2021) and rarely investigate nonlinear relationships, particularly in the context of FS (Wu et al. 2024). Moreover, the literature often neglects cross-country heterogeneity, especially the distinction between

developed and emerging economies (Orlowski 2023; Phan et al. 2021). Prior research has demonstrated that EPU negatively impacts financial stability and innovation, particularly in emerging markets (Desalegn et al. 2023; Jia et al. 2021), while firm-level perceptions of EPU reduce innovation efficiency (Zhou et al. 2023). Governance can mitigate the adverse impact of EPU on FS (Ali et al. 2023).

However, most work assumes a linear relationship and overlooks the possibility that EPU may have threshold effects that could enhance FS by triggering institutional reforms or stronger regulation. This study addresses these gaps by examining the nonlinear relationship between EPU and FS using panel data from 12 countries, divided into six developed and six emerging economies. This setting highlights significant differences in institutional quality, financial development, and macroeconomic resilience, which likely impact how policy uncertainty affects financial stability. The study employs multiple econometric methods, including fixed effects, feasible generalized least squares, dynamic panel GMM, spline regressions, and panel-corrected standard errors, to ensure robustness. By incorporating threshold analysis and heterogeneity across country groups, this research provides new insights into how varying levels of policy uncertainty influence financial stability and offers policy implications for improving systemic resilience.

2.3 | Hypothesis Development

2.3.1 | Nonlinear Relationship Between EPU and Financial Stability

EPU disrupts capital allocation, impairs credit growth, and reduces market confidence, thereby undermining FS (Bordo et al. 2016; Nguyen et al. 2020; Phan et al. 2021). Higher EPU increases market volatility and elevates financial risk, especially in economies with weaker institutional frameworks (Wu et al. 2024). However, the effect of EPU on FS may not follow a simple linear trajectory. Fundamental Options Theory suggests that firms defer investment during uncertainty, initially weakening FS (McDonald and Siegel 1986). Nevertheless, beyond a certain threshold, policymakers and institutions often respond with reforms, interventions, or governance enhancements that restore or even strengthen FS as indicated by Baker et al. (2016) and Xu et al. (2021). Moreover, prior studies note that moderate levels of uncertainty can encourage better risk management, regulatory innovation, and institutional adaptation (Kayani et al. 2024; Singh 2024). This creates the possibility that the EPU–FS relationship may reverse direction once uncertainty exceeds a critical point implying a U-shaped or nonlinear association. Therefore, this study proposes the following hypothesis:

H1. *The relationship between Economic Policy Uncertainty and Financial Stability exhibits a nonlinear pattern, whereby institutional responses may have moderating impacts from very high EPU.*

2.3.2 | Developed vs. Emerging Economies Hypothesis

Although EPU has deleterious effects on FS in many countries, its scale and transmission mechanisms differ in developed and developing countries. Stronger regulatory systems and broader financial markets in developed economies have helped them become more resilient to EPU-induced shocks (Javaheri et al. 2022; Nguyen 2021). Emerging economies, on the other hand, have lower financial institutions, greater economic instability, and increased vulnerability to policy changes, thereby aggravating the negative consequences of EPU (Balcilar et al. 2022; Donadelli and Persha 2014). Furthermore, empirical research suggests that EPU has a more significant and detrimental impact in economies with limited regulatory capital, smaller financial systems, and greater market rivalry (Phan et al. 2021; Seker 2023). Globally, EPU significantly affects local financial markets in the BRIC nations. For instance, spillover effects differ by nation, with China experiencing the most considerable spillover and Brazil the least (Kayani et al. 2024; Smales 2020). This implies that EPU changes affect global uncertainty shocks more likely in emerging markets, thereby compromising financial stability. So, we create the hypothesis:

H2a. *Economic Policy Uncertainty has a more pronounced negative impact on Financial Stability in Emerging Economies than in Developed Economies.*

Moreover, different developed and emerging economies may have different nonlinear effects of EPU on FS. Aman et al. (2024) suggest that whereas regulatory buffers cause less nonlinear effects in developed countries, emerging markets are more sensitive to uncertainty shocks, resulting in greater fluctuations in financial stability. Moreover, EPU affects both developed and developing countries, shifting from negative to positive, although the direction and magnitude are not constant. So, we propose:

H2b. *Emerging economies show a clearer nonlinear link between economic policy uncertainty and financial stability than developed ones due to less financial protection and more policy ambiguity.*

In sum, prior studies primarily examine the linear effects of EPU on financial outcomes Phan et al. (2021) and Das and Ordal (2022) with limited attention to nonlinear relationships, particularly in the context of FS (Wu et al. 2024). Additionally, much of the literature overlooks cross-country heterogeneity, often failing to distinguish between developed and emerging economies (Orlowski 2023; Phan et al. 2021). This study addresses these gaps by investigating the nonlinear impact of EPU on FS, incorporating threshold effects, and conducting a comparative analysis between developed and emerging markets. Specifically, it examines: (i) the nonlinear influence of EPU on FS across country groups, and (ii) how EPU affects different levels of FS low, moderate, and high. Figure 3 presents the conceptual framework that illustrates the relationship between the EPU and FS.

3 | Data and Methodology

3.1 | Data and Sample

We select emerging and developed countries to empirically assess the nexus between EPU and FS. The sample includes six developed (Australia, Canada, Denmark, Japan, United

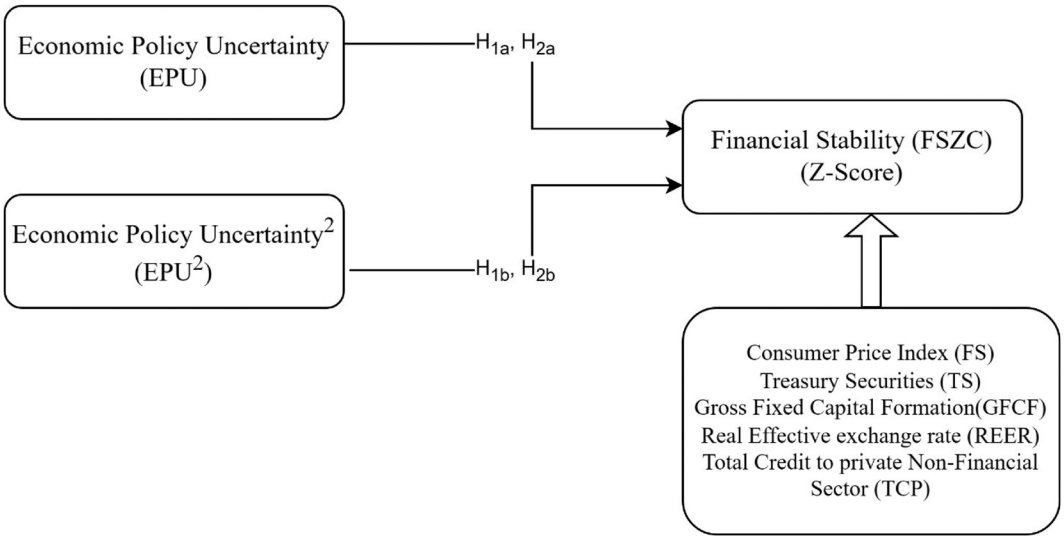


FIGURE 3 | Conceptual framework–economic policy uncertainty and financial stability.

Kingdom, and USA) and six emerging (Brazil, China, India, Mexico, Russia, and South Korea) economies over the period 2003–2022, with a total of 1440 monthly observations from the emerging countries and 1440 monthly observations from the developed countries. The selection of emerging and developed countries is justified by potential differences in the economic development of the two groups of economies and thus, potentially, a difference in the impact of EPU on FS. Data used in this study are collected from various sources, as indicated in Table 1.

3.2 | Variable Description and Measurement

The literature uses various proxies to measure FS (Ang and McKibbin 2007), and no agreement exists on which proxy is superior. This paper uses two proxies widely employed in the academic literature. The first is the banking system country-wise ranking *Z*-score index (FSZS) Sethy and Goyari (2022), which captures the probability of default of a country's banking system.[1] It is given by:

$$\text{Financial Stability}_Z-\text{Score}_{j,t}=\frac{\text{ROA}_{j,t}+\left(\text{Equity}_{j,t}/\text{Assets}_{j,t}\right)}{\sigma\left(\text{ROA}_{j,t}\right)} \tag{1}$$

where ROA indicates return on assets, with assets and returns being country-level aggregate; *j* represents country; *t* indicates time; and σ is standard deviation of the return on total assets. The rationale for using the *Z*-score is that it associates a bank's capital level with variability in its returns, which helps quantify how much variability in returns can be absorbed by the bank's capital without it becoming insolvent. Accordingly, a higher *Z*-score reflects higher banking stability, and vice versa. The second measure of FS is the ratio of M2[2] to GDP, which is used in the robustness check (Broner et al. 2013; Miao and Qamruzzaman 2021; Yanikkaya 2003). The leading independent variable is the EPU index, which encompasses uncertainties related to regulatory, monetary, tax, and spending policies by the official authorities of each selected country. It is computed by Baker et al. (2016) using four components: (1) the frequency of economic policies (FOEP) shown in newspapers; (2) the figure of ended codes; (3) the extent of forecaster disagreement over future inflation (EOFDOFI); and (4) involved government purchase (GP) (Miao and Qamruzzaman 2021; Yanikkaya 2003).

We also incorporate several control variables based on the economic literature, thereby including some economy-wide or country-comprehensive factors that influence FS. Macroeconomic control variables, including the Consumer Price Index (CPI), Treasury Securities (TS), Gross Fixed Capital Formation (GFCF), Real Effective Exchange Rate (REER), and Total Credit to the Private Non-Financial Sector (TCP), are retrieved from FRED. These variables are incorporated to capture key economic and financial dynamics affecting financial stability. The variable, measurement, and empirical references are presented in Table 1.

3.3 | Estimation Model and Empirical Approach

To test our hypotheses and examine the nonlinear association between Economic Policy Uncertainty (EPU) and Financial Stability (FS), we estimate the following baseline model (Benlemlih and Girerd-Potin 2017; Phan et al. 2021; Wu and Chang 2022). Specifically, we set the following baseline model:

$$\begin{aligned}\text{FSZS}_{i,t}=&\alpha_0+\beta_1\text{EPU}_{i,t}+\beta_2\text{EPU}^2{}_{i,t}+{}_1\delta_1\text{CPI}_t+\delta_2\text{TS}_t\\&+\delta_3\text{REER}_t+\delta_4\text{TCP}_t+\delta_5\text{GFCF}_t+\text{V}_1\text{year effect}\\&+\eta_1\text{Country effet}+\varepsilon_{i,t}\end{aligned} \tag{2}$$

where Financial Stability (FS) measured by *Z* score and defined as FSZS, *i* indexes the country, *t* denotes the year, β_1 is the coefficient associated with the economic policy uncertainty index (EPU), and β_2 is the coefficient related to the square of EPU (EPU^2). Consumer price index (CPI), Treasury securities (TS), 3-month rates and yields: Treasury securities (TS), gross fixed

TABLE 1 | Variables definition.

Variables	Syn.	Measurement	Data sources
Financial Stability	FSZS	Country-level or bank-level *Z*-score, capturing the distance to default and insolvency risk based on profitability, leverage, and volatility (Main Analysis)	World Development Indicators (WDI), Fitch Connect
Financial Stability	FS	M2 to GDP ratio, indicating the liquidity position and monetary stability of the economy (Robustness Check)	Federal Reserve Economic Data (FRED)
Economic Policy Uncertainty Index	EPU	EPU Index extracted from newspaper coverage frequency of policy-related economic uncertainty	http://PolicyUncertainty.com
Consumer Price Index	CPI	Standard CPI measure indicating inflation, based on changes in the price level of a market basket of consumer goods and services	FRED
Treasury Securities	TS	3-month interest rates and yields on Treasury securities, serving as a proxy for short-term sovereign borrowing costs	FRED
Gross Fixed Capital Formation	GFCF	GFCF as a percentage of GDP, reflecting net investment in physical assets such as buildings, machinery, and infrastructure	FRED
Real Effective Exchange Rate	REER	Trade-weighted index of a country's currency relative to a basket of other major currencies, adjusted for inflation differentials	FRED
Total Credit to Private Non-Financial Sector	TCP	Total domestic credit extended to households and non-financial corporations, expressed as a share of GDP	FRED

Note: This table provides a variables definition, measurement methods, and data sources.

capital formation (GFCF), real effective exchange rate (REER), and total credit to private non-financial sector (TCP) are control variables. V1 is the year effect, $\eta 1$ represents the country effect, and $\varepsilon_{i,t}$ denotes the error term of the model.

We begin by presenting descriptive statistics and a correlation matrix to understand data patterns and detect potential multicollinearity. Variance Inflation Factor (VIF) analysis is used to confirm multicollinearity diagnostics. Additionally, we test for normality, heteroskedasticity, and autocorrelation using standard diagnostic procedures. Next, we estimate the model using ordinary least squares (OLS) and Fixed Effects (FE) panel regressions. Based on the Hausman test, the fixed effects model is found to be the more appropriate choice. However, diagnostic testing reveals the presence of heteroskedasticity and autocorrelation in the FE model. To address these issues, we apply Feasible Generalized Least Squares (FGLS), which provides efficient and consistent estimates in the presence of such violations (Iftikhar et al. 2025). To verify the nonlinear (U-shaped) relationship between EPU and FS, we adopt the three-step framework proposed by (Haans et al. 2016). In the first step, $\beta 2$ (EPU^2) must be positive and significant. In the second step, the slope at the low end of the EPU range ($\beta 1 + 2 \times \beta 2 \times Xlow$) and slope at the high end of the EPU range ($\beta 1 + 2 \times \beta 2 \times Xhigh$) must be significant. Third, the threshold (Turing) point must be within the range of data. The threshold point is determined as $-\beta_1$ divided by $-2\beta_2$, here β_1 is the coefficient of the linear term (EPU) and β_2 is the square term of EPU^2. To address potential endogeneity and reverse causality between EPU and FS, we use the two-step System Generalized Method of Moments (System GMM) estimator. This technique is suitable for dynamic panel data and effectively controls endogeneity, serial correlation, and omitted variable bias (Arellano and Bond 1991).

We further examine the heterogeneous effect of EPU across different levels of financial stability using quantile regression with 20 bootstrap replications. This method estimates the model at the 25th, 50th, and 75th percentiles of the FSZS distribution, allowing us to explore how the relationship between EPU and FS varies at low, medium, and high levels of stability (Teng et al. 2021). Specifically, we set the following quantile regression model:

$$\begin{aligned} Q(p)_FSZS_{i,t} = {} & \alpha_0 + \beta_1(p) + \beta_2(p) \times EPU_{i,t} + \beta_3(p) \times \left(EPU_{I,t-1}\right)^2 \\ & + \delta_1(p) \times (CPI)_t + \delta_2(p) \times (TS)_t + \delta_3(p) \\ & \times (REER)_t + \delta_4(p) \times (TCP)_t + \delta_5(p) \times (FFCF)_t + \varepsilon_{i,t} \end{aligned} \tag{3}$$

We thank the reviewer for highlighting this important methodological point. In the revised manuscript, we have extended the analysis. In addition to the quadratic model, this is achieved through the application of second-generation non-linear panel estimators. In particular, we have used the Panel Smooth Transition Regression (PSTR) model, as proposed by Gonzalez et al. (2006) to achieve smooth and data-driven regime changes in the effect of Economic Policy Uncertainty (EPU) on Financial Stability (FS). This is because the PSTR framework allows the slope coefficients to be continuously dispersed across regimes of EPU, enabling the testing of non-linearity without a parametric functional structure. As an additional test of the existence of threshold dynamics, we also fitted a Dynamic Panel Threshold Regression (DPTR) model (Kremer et al. 2013) that allows for an endogenous threshold of EPU beyond which its impact on FS reverses. Both non-linear estimators outweigh the substantial effects which are

regime-specific, and these verify the existence of a non-linear long-run relationship as supported by our CS-ARDL findings.

We also perform a robustness check using alternative regression estimators and a substitute measure of FS (FS = M2/GDP). Finally, to ensure further robustness of our findings, we conduct several additional tests. These include applying alternative regression techniques and replacing the primary proxy for financial stability with an alternative measure, specifically the M2-to-GDP ratio. To further validate this nonlinear relationship, we implement spline regression analysis. Spline regression is a flexible technique, beneficial for capturing threshold effects and smooth nonlinearities that are not fully modeled by quadratic terms.

4 | Empirical Results

4.1 | Descriptive Statistics

Table 2 gives the statistical features of the variables used in the study. We report the mean, standard deviation (SD), maximum, and minimum values of both panel variables. We note that the mean value is positive for all variables. The average FSZS by country for the developed countries is 17.70 in conjunction with an SD of 8.058, whereas the average FSZS for emerging countries is 15.77, and the SD is 5.55. Higher banking stability is expected in developed than in emerging economies. The average mean EPU value for developed countries is 132.8, along with an SD of 79.006, whereas the EPU index of emerging countries is 148.096 with an SD of 134.11. Taken together, the mean value of the EPU index in emerging economies is higher than that in developed countries, while the FSZS of developed countries is lower than that of emerging countries. We report the average, SD, maximum, and minimum values of the control's variables.

4.2 | Results From Correlation Matrix

We employ two approaches to assess multicollinearity: the pairwise correlation method, which examines relationships between individual explanatory variables, and the variance inflation factor (VIF) method, which evaluates the overall collinearity among independent variables (Glaeser et al. 2004). As presented in Table 3, the pairwise correlation coefficients fall within an acceptable range, indicating that there is no strong linear dependence among the variables. Similarly, the VIF values remain below 3, while the reciprocal (1/VIF) stays above the critical threshold of 1, confirming that collinearity does not distort the estimation. These findings collectively indicate that multicollinearity is not a concern across any panels, whether for emerging, developed, or the full sample.

4.3 | Econometric Diagnostics and Estimator Selection

4.3.1 | Unit Root Tests (Stationarity)

To assess the stationarity of the panel variables, we conduct the Levin–Lin–Chu (LLC), Pesaran–Shin (IPS), and ADF–Fisher chi-square tests. Most key variables including FS, EPU, and EPU^2 are stationary at level. Control variables such as CPI, REER, and GFCF achieve stationarity after first differencing. These results justify the use of GMM estimators for dynamic modeling and FGLS for robustness checks under heteroskedastic and autocorrelated error structures.

4.3.2 | Pre-Estimation Diagnostics

To ensure the appropriate choice of estimators, we perform a series of pre-estimation diagnostic tests to assess the validity of underlying model assumptions. White's test, the Modified Wald test, and the Wooldridge test for autocorrelation all reject their respective null hypotheses, indicating the presence of heteroskedasticity and serial correlation in the panel data. These violations limit the reliability of OLS and fixed effects estimators under conventional assumptions. The Hausman test further rejects the null of no systematic difference between fixed and random effects, supporting the use of fixed effects models. In addition, the Cameron and Trivedi decomposition indicates non-normality in the residuals; however, this does not invalidate the estimation, as robust standard errors are used. Full diagnostic results are presented in Tables A1 and A2.

4.3.3 | CD Tests (Cross-Sectional Dependence)

Before estimating the long-run relationship between EPU and FSZS, we use several tests to check for cross-sectional dependence (Wang and Taghizadeh-Hesary 2023; Yuen and Yuen 2024). Specifically, we conduct the Breusch-Pagan LM, Pesaran scaled LM, bias-corrected scaled LM, and Pesaran CD tests. Table A3 shows that all tests reject the null hypothesis of no cross-sectional dependence. This means countries in the panel are highly connected, possibly due to global financial links and similar economic movements. To address this, we employ the Fully Modified Ordinary Least Squares (FMOLS) method, which adjusts for cross-sectional dependence, serial correlation, and endogeneity, thereby providing more reliable long-run estimates.

4.3.4 | Panel Co-Integration Test and CS-ARDL Estimation

In this section, we apply the Pedroni (2004), Kao (1999) and Westerlund (2008) panel cointegration test to examine the long-run equilibrium relationship between EPU and FSZS, following the approaches of. The evidence of panel cointegration is strong, with both second-generation and first-generation procedures. All tests reject the null hypothesis of a non-existent long-run relationship in both linear and nonlinear specifications (see Table A4). This facilitates the stable balance relation between EPU (and EPU^2) and financial stability.

Following the results of the Cointegration test, the CS-ARDL results in Table A5 show that Economic Policy Uncertainty (EPU) and Financial Stability (FS) have a strong and statistically significant long-term relationship. Higher EPU always hurts FS in the linear specification; however, in the nonlinear

TABLE 2 | Descriptive statistics.

Variables	Obs	Mean	Std. dev.	Min	Max
Emerging countries					
FSZS	1440	15.78	5.56	6.22	29.76
FS	1440	55.44	141.84	0.00	675.73
EPU	1440	148.10	134.11	8.51	970.83
Log_EPU	1440	4.88	0.72	2.14	6.88
CPI	1440	83.91	22.50	30.89	132.10
TS	1440	6.15	3.71	0.00	28.78
REER	1440	100.69	16.74	62.66	150.85
TCP	1440	411,000	927,000	44.70	4,020,000
GFCF	1440	28.26	2.25	24.97	32.64
Developed countries					
FSZS	1440	17.70	8.06	4.96	35.12
FS	1440	30.35	45.73	0.20	172.35
EPU	1440	132.80	79.01	24.04	678.82
Log_EPU	1440	4.85	0.60	3.18	6.52
CPI	1440	80.83	37.05	−1.92	116.17
TS	1440	3.31	4.55	−1.72	21.75
REER	1440	100.73	16.56	65.20	156.64
TCP	1440	149,000	316,000	706.48	982,000
GFCF	1440	12.00	1.25	9.32	13.30
Full sample					
FSZS	2880	16.74	7.31	4.96	35.12
FS	2880	42.90	100.42	0.00	675.73
EPU	2880	140.45	106.46	8.51	970.83
Log_EPU	2880	4.78	0.69	2.14	6.88
CPI	2880	82.37	29.37	−1.92	132.10
TS	2880	4.73	4.13	−1.72	28.78
REER	2880	100.71	16.66	62.66	156.64
TCP	2880	280,000	621,500	44.70	4,020,000
GFCF	2880	20.00	7.00	1504.00	11174.13

Note: This table gives the descriptive statistics for emerging and developed countries and the full panel. Economic policy uncertainty (EPU) is the thematic independent variable. Financial stability country *Z*-score (FSZS) by country is the dependent variable. Control variables are consumer price index (CPI), Treasury securities (TS), 3-month rate of Treasury securities (TS), gross fixed capital formation (GFCF), real effective exchange rate (REER), and total credit to the private non-financial sector (TCP).

model, uncertainty has a U-shaped effect on financial stability: initially, it decreases, but then it increases more strongly than the uncertainty, indicating adaptive responses by regulatory authorities and the market to the uncertainty. This asymmetry is supported by short-run dynamics, which reveal an immediate negative impact of EPU that dissipates with an increase in uncertainty. The control variables have stabilizing and destabilizing effects on inflation (CPI) and exchange-rate misalignments (REER), as well as trade openness (TS), credit depth (TCP), and capital formation (GFCF), respectively. The negative and substantial error correction term (ECT = −0.42) indicates a high convergence rate per month, with approximately 42% of the disequilibrium being corrected in every period. On the whole, the results suggest the critical role of policy uncertainty in the short and long-run processes of resilience and adjustment of the financial system.

TABLE 3 | Pairwise correlation.

Panel A: emerging countries								
Variables	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**VIF**	**1/VIF**
(1) EPU	1.000						1.14	0.8
(2) CPI	0.278***	1.000					1.25	0.8
(3) TS	0.042	−0.274***	1.000				1.42	0.7
(4) REER	−0.039	0.011	0.217***	1.000			1.16	0.86
(5) TCP	0.019	0.246***	−0.414***	−0.067**	1.000		2.96	0.34
(6) GFCF	0.001	0.240***	−0.422***	−0.059**	0.993***	1.000	1.02	0.98
Mean VIF							1.49	
Panel B: developed countries								
Variables	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**VIF**	**1/VIF**
(1) EPU	1.000						1.26	0.79
(2) CPI	0.050*	1.000					1.39	0.72
(3) TS	−0.176***	0.256***	1.000				1.03	0.97
(4) REER	−0.045	0.129***	−0.037	1.000			1.38	0.72
(5) TCP	−0.141***	0.189***	−0.336***	0.658***	1.000		1.65	0.61
(6) GFCF	−0.081***	0.535***	−0.365***	0.085***	0.521***	1.000	1.66	0.6
Mean VIF							1.39	
Panel C: overall sample								
Variables	**(1)**	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**VIF**	**1/VIF**
(1) EPU	1.000						1.94	0.54
(2) CPI	0.149***	1.000					1.25	0.8
(3) TS	−0.071***	0.091***	1.000				1.42	0.70
(4) REER	−0.040**	0.081***	0.074***	1.000			1.11	0.90
(5) TCP	0.005	0.179***	−0.254***	0.102***	1.000		1.36	0.73
(6) GFCF	0.024	0.134***	−0.142***	−0.040**	0.934***	1.000	1.01	0.99
Mean VIF							1.16	

Note: *, **, and *** indicate significance at $p < 0.1$, $p < 0.05$, and $p < 0.01$, respectively. This table provides the correlations and VIF estimates for emerging and developed countries, respectively.

4.4 | Baseline Results

4.4.1 | Estimation From OLS and FE Models

We test the effect of EPU on FS across emerging markets, developed economies, and the full sample in Table 4. The regressions include both linear and squared terms of EPU to capture possible nonlinear effects. Results from ordinary least squares and fixed effects models show that EPU negatively affects FS, while the squared term shows a positive and significant coefficient. This confirms a nonlinear (U-shaped) relationship across all subsamples. In the case of emerging markets, EPU exhibits a more substantial and consistently adverse effect, revealing greater sensitivity to uncertainty. The turning point occurs at a lower threshold for emerging economies, signaling earlier institutional strain and a more proactive policy response. Developed markets exhibit similar patterns but with weaker magnitudes. These findings suggest that while EPU initially erodes FS, it may trigger policy adaptation or lead to institutional strength once it reaches a certain threshold, especially in emerging markets.

4.4.2 | Estimation From Feasible Generalized Least Squares Regression

We apply the Feasible Generalized Least Squares (FGLS) regression to correct for panel-specific heteroskedasticity and serial correlation, as confirmed by diagnostic tests in Tables A1 and A2. Table 5 presents the FGLS results, which show that EPU has an adverse effect on FS, while the squared term of EPU (EPU^2) reveals a nonlinear pattern. The positive coefficient on EPU^2 suggests that after a certain level, the negative impact of EPU starts to weaken. This implies that financial systems build resilience or that only high levels of uncertainty lead to further instability.

The overall results confirm a nonlinear relationship, as FS does not react uniformly to shifts in EPU. In emerging markets, weak institutional frameworks and heightened vulnerability to global shocks exacerbate the adverse effects. However, developed economies handle uncertainty more effectively, which helps reduce the negative impact once EPU rises beyond a certain threshold.

The inclusion of the squared EPU term follows a quadratic specification. The turning point is calculated using the formula $x^* = \frac{\beta_1}{2\beta_2}$, which identifies the level of EPU at which the direction of its effect on FS changes. The slope at the minimum and maximum observed values of EPU is given by $\beta_1 + 2\beta_2 \times \text{EPU}$, which helps assess whether the relationship increases or decreases across the range. Besides, the Sasabuchi and Wald tests confirm a U-shaped relationship, with significant linear and squared terms and valid threshold values within the EPU range (Nguyen 2021; Phan et al. 2021). These results support the relevance of the inflection point and align with prior studies, highlighting the need to consider the nonlinear effects of EPU on FS.

4.4.3 | Visualizing the Threshold Effects of EPU on FSZS Across Country Groups

Figure 4 shows the nonlinear relationship between EPU and FSZS across emerging markets, developed economies, and the full sample. Each curve follows a quadratic form, which reflects how FSZS shifts at different levels of EPU. In emerging markets, FSZS falls as EPU rises, reaching its minimum at an EPU value near 4.69. After this point, FSZS rises, possibly due to institutional reforms or stronger policy responses. Developed markets also display a U-shaped curve, but with a higher threshold of 9.63, which suggests that these economies absorb more uncertainty before FSZS begins to improve. The full sample shows a lower turning point at 2.31, which may reflect aggregated effects or quicker responses by some countries. Each curve identifies the level at which the marginal effect of EPU on FSZS turns positive. Figure 5 presents the FGLS regression coefficients for EPU and EPU^2. The results confirm that EPU first reduces FS, then raises it after a threshold.

TABLE 4 | Baseline regression results.

	Emerging countries		Developed countries		Full sample	
	(1)	(2)	(3)	(4)	(5)	(6)
Variables	**FSZS**	**FSZS**	**FSZS**	**FSZS**	**FSZS**	**FSZS**
EPU	−0.0291***	−0.0055***	−0.0364***	−0.0185***	−0.0833***	−0.0398***
	(0.0008)	(0.0029)	(0.0139)	(0.0035)	(0.0315)	(0.0009)
EPU2	0.0806	0.4805***	0.8605***	1.4608	0.0233***	0.0114***
	(0.4006)	(0.0016)	(0.0606)	(1.0306)	(0.0021)	(0.0301)
CPI	0.0960***	0.0678***	−0.0345***	−0.173***	0.206***	0.0178
	(0.0048)	(0.0045)	(0.00428)	(0.0209)	(0.0370)	(0.0381)
TS	0.6605	−0.0713**	−0.234	0.173***	−3.444***	−0.681**
	(0.6507)	(0.0312)	(0.4325)	(0.0470)	(0.285)	(0.322)
REER	0.0919	0.0592***	−0.0734***	−0.103***	−1.106***	−0.281***
	(0.0951)	(0.0058)	(0.0095)	(0.00848)	(0.0749)	(0.0738)
TCP	7.0106***	−5.0707	4.1106***	−1.2005***	−0.192	−0.0219
	(0.4007)	(9.3507)	(6.1807)	(0.316)	(0.222)	(0.179)
GFCF	−3.100***	0.1003	6.8506***	8.1515***	9.8701	0.9901***
	(0.2100)	(0.2001)	(9.6007)	(4.1506)	(5.1211)	(0.0311)
Constant	3.039***	5.329***	2.49	19.71***	−9.0121***	−5.0871***
	(0.643)	(0.788)	(4.019)	(1.4900)	(0.0012)	(0.0004)
Year FE		Yes		Yes		Yes
Country FE		Yes		Yes		Yes
Observations	1440	1440	1440	1440	2880	2880
R-squared	0.207	0.201	0.263	0.211	0.291	0.244
Number of ids	6	6	6	6	12	12

Note: This table reports the classic ordinary least squares and fixed effects results for emerging and developed countries. Standard errors are in parentheses. *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

TABLE 5 | Feasible generalized least squares regression estimates.

	Emerging countries	Developed countries	Full sample
	(1)	(2)	(3)
Variables	**FSZS**	**FSZS**	**FSZS**
EPU	−0.0319***	−0.0289***	0.0282***
	(0.0026)	(0.0048)	(0.0055)
EPU^2	0.0034***	0.0015***	0.0061***
	(0.0012)	(0.0001)	(0.0005)
CPI	0.0900***	−0.1347***	−0.1005
	(0.0062)	(0.0081)	(0.0917)
TS	0.1097***	−0.4330***	−6.742***
	(0.0409)	(0.0407)	(0.601)
REER	−0.0479***	0.0119***	−1.055***
	(0.0018)	(0.0045)	(0.118)
TCP	−0.0143	0.0567*	−0.257
	(0.0222)	(0.0357)	(0.367)
GFCF	−0.8966***	−0.9126***	−0.832***
	(0.0624)	(0.0044)	(0.0080)
Constant	1.0503***	7.6821***	−0.0112***
	(0.6370)	(0.1525)	(0.0001)
Obs.	1440	1440	2880
R-squared	0.2169	0.2515	0.3761
Wald Chi^2	153.89	216.54	453.18
Slope at the low end of X-range	0.026	0.043	0.0757
Slope at the high end of X-range	6.57	2.01	11.83
Sasabuchi test (statistic)	$t_{min} = 1.26$ $t_{max} = 2.82$	$t_{min} = 6.36$ $t_{max} = 14.78$	$t_{min} = 1.50$ $t_{max} = 12.16$
Threshold point [x*]	4.69	9.63	2.31
EPU	8.51–970.83	24.04–678.82	8.51–970.83
Log_EPU	2.14–6.88	3.18–6.52	2.14–6.88
Threshold/within data range?	Yes	Yes	Yes

Note: This table reports feasible generalized least square (FGLS) regression results for emerging markets, developed countries, and the full sample. The dependent variable is the Financial Stability *Z*-score (FSZS). EPU and its squared term capture the nonlinear effect of policy uncertainty. The threshold point is given by the formula $x^* = \frac{\beta_1}{2\beta_2}$. Slopes at the low and high ends of EPU assess marginal effects. The Sasabuchi test confirms the U-shaped relationship. All thresholds lie within the observed EPU range. Standard errors are in parentheses. *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.

4.4.4 | FMOLS Estimation for EPU-FS Long-Run Relationship

Given the evidence of cointegration, we use Fully Modified Ordinary Least Squares (FMOLS) to estimate the long-run relationship between EPU and FSZS. FMOLS corrects for serial correlation and endogeneity, providing reliable coefficient estimates. We include both EPU and its squared term (EPU^2) to capture nonlinear effects. Table 6 shows a nonlinear (U-shaped) long-run relationship between EPU and FSZS. In emerging markets, EPU reduces FSZS up to a threshold, after which the effect becomes positive, which suggests institutional adjustments under rising uncertainty. Developed economies exhibit a similar pattern, but with a higher threshold, reflecting stronger institutional capacity. These results support real options theory and financial cycle theory, which suggest that uncertainty initially suppresses financial outcomes but may lead to long-term improvements through policy reform and institutional responses. The findings underscore the necessity for regulatory strategies tailored to distinct market contexts to ensure financial stability in uncertain policy environments.

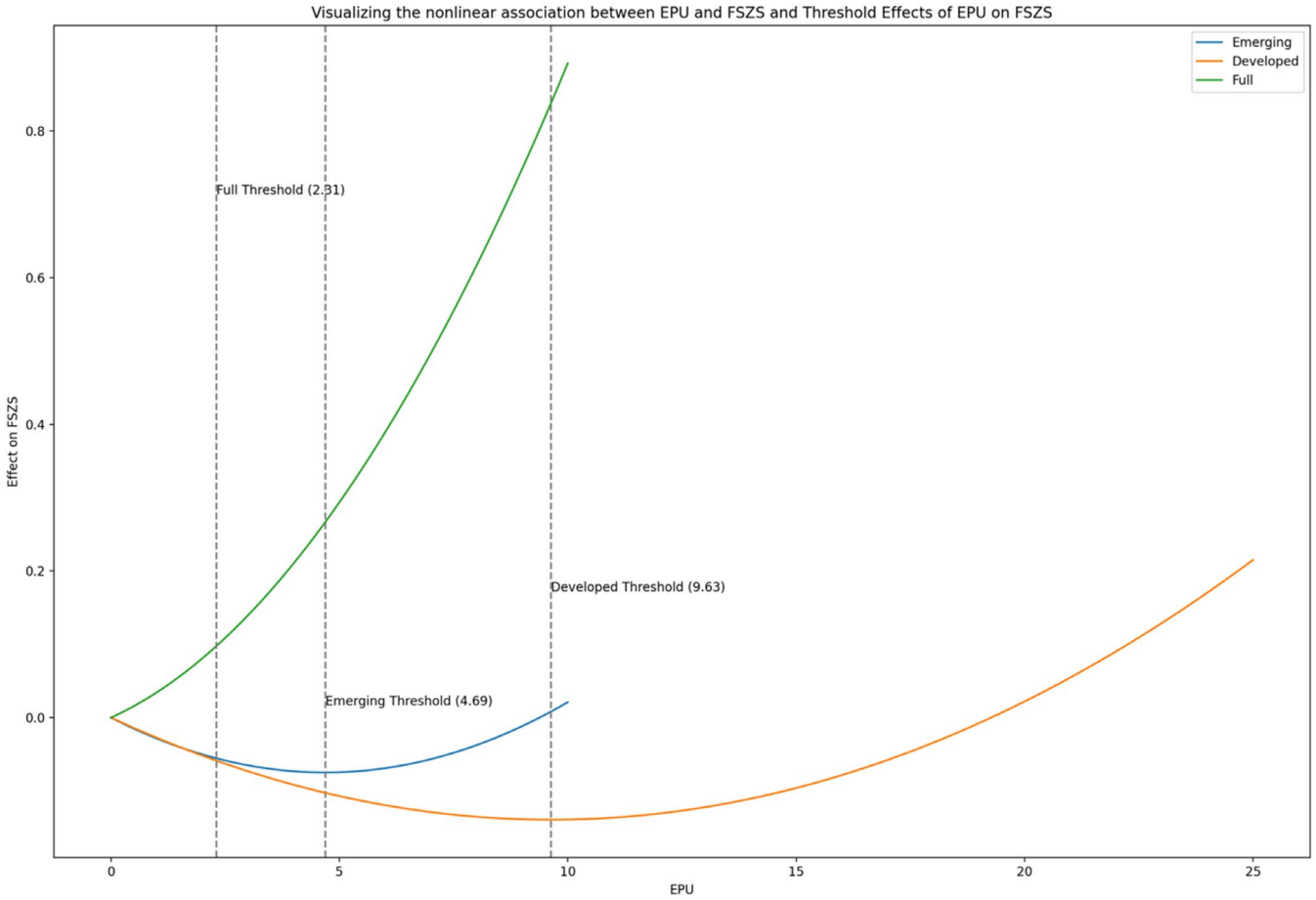


FIGURE 4 | Threshold effects of EPU on FSZS across country groups.

4.5 | Endogeneity and Distributional Heterogeneity

4.5.1 | Dynamic Panel Estimation to Address Endogeneity

The baseline estimations may suffer from endogeneity due to reverse causality, omitted variable bias, and unobserved heterogeneity—common issues in panel data analysis. To address these concerns, we employ the two-step System Generalized Method of Moments (System GMM) estimator, which is appropriate for dynamic panels with endogenous regressors (Bordo et al. 2016; Wu et al. 2024). Table 7 presents the GMM results. The coefficient on EPU is negative and significant, suggesting that higher policy uncertainty reduces financial stability. The positive and significant squared term (EPU^2) confirms a non-linear relationship, implying that beyond a certain threshold, increases in EPU may improve FS. The turning point varies across developed and emerging economies. The inclusion of lagged dependent variables introduces dynamic panel bias, which traditional estimators cannot correct. System GMM overcomes this by using internal instruments derived from lagged values. The Arellano–Bond test confirms the absence of second-order autocorrelation. While the Sargan test indicates possible over identification, the Hansen test supports instrument validity, confirming robust and consistent estimates.

4.5.2 | Quantile Regression Estimation for Distributional Heterogeneity

We report the results of the quantile regression in Table 8 for high, moderate, and low levels of FS. The findings offer further granularity, revealing that the negative impact of EPU on FS is more pronounced at higher levels of FS. This suggests that financially stable economies may be more susceptible to policy uncertainty, as they rely more on structured financial frameworks that are disrupted by unpredictable policy shifts. The differential effects observed between emerging and developed economies reinforce the notion that institutional maturity and market depth play critical roles in mitigating uncertainty shocks. The estimated results indicate that EPU has a negative impact on high levels of FS in the full sample. When evaluating emerging and developed countries separately, we find that in emerging economies, the negative impact of EPU on FS intensifies across the distribution. In contrast, in developed economies, EPU has a positive effect on moderate levels of FS. Regarding the control variables, CPI and GFCF exhibit a positive impact, whereas TS, REER, and TCP have a negative effect on FS.

4.5.3 | Panel Smooth Transition Regression (PSTR) and Dynamic Panel Threshold Regression (DPTR)

The findings of the Panel Smooth Transition Regression (PSTR) and Dynamic Panel Threshold Regression (DPTR) models, presented in Table 9, all indicate the existence of a statistically

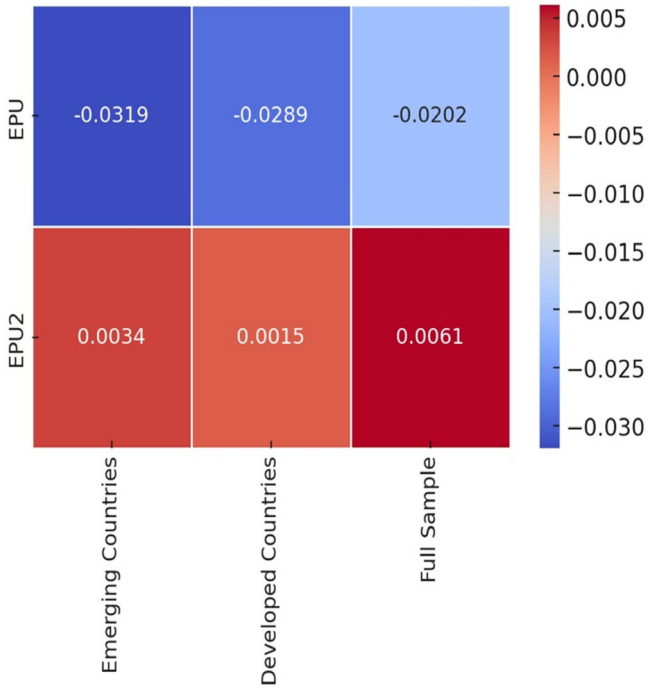


FIGURE 5 | Visual representation of the GLS regression estimation results.

significant nonlinear correlation between Economic Policy Uncertainty (EPU) and Financial Stability (FS). Both methods identify two distinct regimes: low and high uncertainty, separated by a threshold in the EPU variable, which is endogenously estimated (approximately 4.7 in log form).

As seen in the transition function, in the PSTR model, the marginal effect EPU has on FS gradually increases. At reduced uncertainty levels, EPU has a negative impact on financial stability, which suggests that average policy uncertainty will erode market confidence and weaken the financial system. Nevertheless, beyond the estimated level, this adverse effect decreases and turns out to be positive, implying that the financial system adjusts to an increased level of uncertainty. The estimated slope parameter confirms the presence of a relatively steep and smooth transition, indicating regime-specific resilience in financial markets.

Results of the DPTR support this nonlinearity evidence but show a more abrupt shift in regimes which is discrete. The effect of EPU, below the estimated threshold, decreases FS; however, its marginal effect is positive above the threshold, suggesting that extreme uncertainty triggers corrective responses, including stricter regulatory supervision or reduced risk-taking, which stabilizes the system. The presence of the lagged FS term ensures the maturity of persistence in financial stability. In contrast, the presence of a substantive and negative error-correction term confirms the long-run adjustment to the equilibrium.

In both models, the control variables act as expected: the weakening of FS is caused by inflation (CPI) and exchange-rate volatility (REER); trade openness (TS), credit depth (TCP), and capital formation (GFCF) contribute to a stronger FS. In general, the evidence presented in the article on PSTR and DPTR together is quite convincing to uphold the hypothesis of having a nonlinear, threshold-dependent relationship between EPU and FS with respect to which the financial stability initially depreciates with the increase in uncertainty. However, the opposite holds after the uncertainty reaches a critical threshold, which demonstrates the adaptability and self-correcting quality of the financial system.

TABLE 6 | FMOLS regression for long-run relationship between EPU and FSZS.

	Emerging countries	Developed countries	Full sample
	(1)	(2)	(3)
Variables	**FSZS**	**FSZS**	**FSZS**
EPU	−0.0381***	−0.0217***	−0.0297***
	(0.0098)	(0.0168)	(0.0066)
EPU^2	1.0467 ***	0.8424***	0.5287***
	(0.1528)	(0.2604)	(0.1069)
Constant	−1.0205***	−5.0432***	21.524***
	(0.0040)	(1.0921)	(1.275)
Controls	Yes	Yes	Yes
Obs.	1440	1440	2880
R-squared	0.8629	0.8368	0.9780
Adjusted *R*-squared	0.8619	0.8351	0.9778
SE of regression	23.8393	31.5937	6.9538
Mean dependent variable	29.5384	27.6730	31.3963
SD dependent variable	64.1429	77.8133	46.6696
Sum squared residuals	1,382,138	1,207,773	58752.2
Long-run variance	167.351	302.139	26.986

Note: This table reports Fully Modified Ordinary Least Squares (FMOLS) estimates of the long-run relationship between EPU and FSZS, including a nonlinear term (EPU^2). FMOLS corrects for serial correlation and endogeneity. Standard errors are in parentheses. ***$p < 0.01$.

4.6 | Robustness Check

To ensure the reliability and consistency of the baseline results, the study conducts a series of robustness checks using alternative measures of financial stability and different econometric estimation techniques.

4.6.1 | Alternative Measures and Estimation Techniques

First, to strengthen the credibility of our baseline results, we rerun the core models using a set of alternative estimation

TABLE 7 | Dynamic panel estimation using two-step GMM.

	Emerging countries	Developed countries	Full sample
	(1)	(2)	(3)
Variables	**FSZS**	**FSZS**	**FSZS**
L.FSZS	−0.4601***	−0.0118***	−0.1801***
	(0.1660)	(0.0038)	(0.0145)
EPU	−0.1707**	−0.0105***	−0.0209***
	(0.0744)	(0.013)	(0.0012)
EPU^2	0.0015***	−3.0321***	−2.0010***
	(0.0001)	0.6450	(0.5000)
CPI	0.0211***	0.676	1.735***
	(0.0010)	(0.445)	(0.903)
TS	0.0027***	0.2309***	−4.947***
	(0.0010)	(0.0767)	(0.9803)
REER	−0.0989	1.2301	6.3404
	(0.0963)	(0.0311)	(0.5216)
TSP	0.0000	2.312***	−9.8008***
	(0.0100)	(0.0320)	(0.3011)
TCP	1.831***	−2.171	−0.0173***
	(0.0012)	(2.2601)	(0.0003)
GFCF	−0.0303***	0.0337	0.3411***
	(0.0013)	(1.321)	(0.0211)
Constant	1.0031	2.3011***	−9.0007***
	(0.0013)	(0.0231)	(1.588)
Observations	1324	1367	2731
Slope at the low end of X-range	0.085	0.7296	0.7402
Slope at the high end of X-range	2.7418	80.75	83.72
Sasabuchi test (statistic)	$t_{min} = 2.65$ $t_{max} = 6.84$	$t_{min} = 4.91$ $t_{max} = 9.17$	$t_{min} = 5.14$ $t_{max} = 10.93$
Threshold Point [x*]	56.90	35.21	34.32
EPU	(8.51–970.83)	(24.04–678.82)	(8.51–970.83)
Log_EPU	(2.14–6.88)	(3.18–6.52)	(2.14–6.88)
Threshold/within data range?	Yes	Yes	Yes
Arellano-Bond test for AR (1)	0.0000	0.0000	0.0000
Arellano-Bond test for AR (2)	0.2101	0.2133	0.2711
Sargan test of over identification	0.1108	0.1923	0.1061
Hansen test	0.2101	0.2103	0.2015
Wald Chi^2	237.01***	104.09***	6901.49***

Note: This table presents the GMM estimation for emerging and developed countries and the full sample. The Sargan–Hansen test examines the bounds of over identification. The instruments are valid and independent of the error term. Missing instruments are correctly ruled out of the equation; this is the joint null hypothesis. Parenthetical robust standard errors at $*p = 0.1$, $**p = 0.05$, $***p = 0.01$.

TABLE 8 | Impact of EPU on high, moderate, and low levels of financial stability.

FSZS-regimes	High FSZS	Moderate FSZS	Low FSZS	High FSZS	Moderate FSZS	Low FSZS	High FSZS	Moderate FSZS	Low FSZS
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
	FSZS	FSZS	FSZS	FSZS	FSZS	FSZS	FSZS	FSZS	FSZS
Variables	**Emerging countries**			**Developed countries**			**Full sample**		
EPU	−0.0148**	0.0071	0.0081	−0.0047	0.0191***	0.0601	−0.0729***	0.2741	0.1841
	(0.0020)	(0.0119)	(0.0081)	(0.0015)	(0.0066)	(0.0521)	(0.0015)	(0.2021)	(0.2030)
EPU^2	0.2901***	−1.0601***	−1.851	0.6406	−3.1305**	−0.0010	0.0009***	−0.0011	−0.0107
	(0.0212)	(0.6706)	(1.401)	(9.106)	(1.5401)	(9.0705)	(0.0007)	(030012)	(0.2032)
CPI	−0.0599***	−0.0647***	−0.0658***	−0.0272***	−0.0149**	−0.497*	0.3707***	0.16962***	0.0435
	(0.0091)	(0.0055)	(0.0169)	(0.0084)	(0.0063)	(0.274)	(0.0907)	(0.0219)	(0.0335)
TS	−0.305***	−0.0484	−0.105**	−0.532***	−0.811***	−8.283***	−3.5444***	−2.3522***	−1.2231***
	(0.0426)	(0.0546)	(0.0507)	(0.0832)	(0.157)	(0.479)	(0.7393)	(0.1808)	(0.2277)
REER	−0.0205**	−0.133***	−0.262***	−0.0450***	−0.0602***	−1.642***	−1.2253***	−0.0718	0.1060**
	(0.0103)	(0.0246)	(0.0784)	(0.0173)	(0.0073)	(0.152)	(0.2605)	(0.0444)	(0.0466)
GFCF	−0.0347**	−0.0354**	−0.0480	−0.0467***	−0.0487	−0.384	0.2507***	0.0093***	0.0531
	(0.0148)	(0.0178)	(0.0536)	(0.0146)	(0.0372)	(0.486)	(0.0801)	(0.0009)	(0.6039)
TCP	2.6301***	2.7701***	2.5505***	1.9905***	1.7005***	−2.1605**	−0.0032***	−0.0036***	(0.0005) ***
	(1.8101)	(5.2201)	(7.6207)	(3.2307)	(1.4306)	(8.4006)	(0.0001)	(0.0003)	(0.0001)
Constant	−0.6512***	−0.0043***	−0.0145***	−0.0934***	−1.0205***	−5.0432	21.524***	4.8322	−8.4819
	(0.0002)	(0.00001)	(0.0011)	(0.0002)	(0.0040)	(5.0921)	(1.275)	(18.1504)	(10.9845)
Number of id	6	6	6	6	6	6	12	12	12
Pseudo R^2	0.302	0.187	0.121	0.213	0.191	0.291	0.294	0.141	0.213

Note: This table provides the estimated results of the impact of EPU on high, moderate, and low levels of financial stability in emerging and developed countries. The dependent variable is *Z*-score by country (FSZS). Standard errors are in parentheses, $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

TABLE 9 | Panel smooth transition regression (PSTR) and dynamic panel threshold regression (DPTR) results.

	PSTR	DPTR
Variable	**Low-EPU regime (β_1)**	**Below threshold (EPU ≤ τ)**
Lagged FS ($t-1$)		0.432***
		(0.064)
EPU	−0.231***	−0.224***
	(0.062)	(0.057)
EPU^2		0.028**
		(0.013)
CPI	−0.078**	−0.082**
	(0.033)	(0.039)
TS	0.114**	0.118**
	(0.046)	(0.050)
REER	−0.055**	−0.059**
	(0.026)	(0.027)
TCP	0.171***	0.176***
	(0.039)	(0.040)
GFCF	0.089***	0.097***
	(0.022)	(0.023)
Error-correction term (ECT_{t-1})		−0.411***
		(0.082)
	High-EPU regime (β_2)	**Above threshold (EPU > τ)**
Lagged FS ($t-1$)		0.418***
		(0.065)
EPU	0.087*	0.092**
	(0.049)	(0.046)
CPI	−0.066**	−0.069*
	(0.031)	(0.037)
TS	0.096**	0.103**
	(0.043)	(0.049)
REER	−0.042*	−0.047**
	(0.024)	(0.026)
TCP	0.164***	0.163***
	(0.038)	(0.039)
GFCF	0.091***	0.093***
	(0.023)	(0.024)

(Continues)

TABLE 9 | (Continued)

	High-EPU regime (β_2)	Above threshold (EPU > τ)
Error-correction term (ECT_{t-1})		−0.426***
		(0.080)
Difference ($\beta_2-\beta_1$)	0.318***	
	(0.081)	
Slope parameter (γ)	8.73***	
	(2.17)	
Threshold (c)	4.68 (log-EPU index)	
Threshold (τ)		4.73 (log-EPU)
Observations	2880	2880
Countries (N)	12	12
Adjusted R^2	0.84	0.83

Note: In this table, Robust standard errors are reported in parentheses. Significance levels are denoted as *$p < 0.10$, **$p < 0.05$, and ***$p < 0.01$. In Column 1 (PSTR model), the estimated threshold ($c \approx 4.68$) indicates that when EPU exceeds this value, its marginal effect on financial stability becomes positive, confirming a smooth, regime-dependent nonlinearity. The relatively high slope parameter (γ) suggests a sharp yet continuous transition between low- and high-uncertainty regimes. In Column 2 (DPTR model), the threshold parameter (τ) and corresponding standard errors are obtained through 1000 bootstrap replications. All estimated coefficients are robust to heteroskedasticity and cross-sectional dependence. The significance of both EPU and EPU^2 terms further validates the presence of a nonlinear, threshold-dependent long-run relationship between economic policy uncertainty and financial stability. For PSTR: Transition variable = EPU; Transition function = logistic ($m = 1$). For DPTR: Dependent variable = Financial Stability (FS); Threshold variable = EPU; Estimator = System GMM with bootstrap threshold search.

techniques and proxy measures for financial stability. These robustness checks are conducted to ensure that model-specific assumptions or measurement limitations do not drive the observed effects of EPU. Table 10 presents the outcomes of three complementary robustness approaches. In Columns 1–3, we re-estimate the baseline models using the Feasible Generalized Least Squares (FGLS) estimator, which accounts for heteroskedasticity and autocorrelation in panel data, with the FSZS as the dependent variable. In Columns 4–6, we employ an alternative proxy for financial stability, broad money (M2), as a percentage of GDP, to test whether our results hold under a different conceptualization of systemic soundness. Finally, Columns 7–9 apply Panel-Corrected Standard Errors (PCSE), which allow for cross-sectional dependence and panel-specific heteroskedasticity, again using FSZS as the outcome variable. Across all estimations, the coefficient on EPU remains negative and statistically significant, confirming its destabilizing effect. The squared term of EPU (EPU^2) consistently shows a positive and significant influence on FS, reinforcing the presence of a nonlinear U-shaped relationship. The direction and significance of the control variables remain broadly consistent, further validating the robustness of the findings.

TABLE 10 | Robustness check with alternative estimators and measures.

	Alternative estimator (FGLS)			Alternative measure of FS (M2 to GDP)			Panel-corrected standard errors (PCSE)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Developed	Emerging	Full sample	Developed	Emerging	Full sample	Developed	Emerging	Full sample
Variables	FSZS	FSZS	FSZS	FS	FS	FS	FSZS	FSZS	FSZS
EPU	−0.0167***	−0.0216***	−0.0305***	−0.0341***	−0.1805***	−0.0204***	−0.0113***	−0.0805**	−0.0147***
	(0.0031)	(0.0020)	(0.0150)	(0.0161)	(0.0450)	(0.0051)	(0.0047)	(0.0042)	(0.0092)
EPU^2	0.0112***	0.0031***	0.0016***	0.0019**	0.0006***	0.0319***	0.4607***	0.7906***	0.5112***
	(0.0001)	(0.0001)	(0.0018)	(0.0004)	(0.0008)	(0.0104)	(0.0007)	(0.0016)	(0.0501)
CPI	−0.1653***	0.0751***	0.0401***	−0.01902***	0.2479***	0.1055***	0.0891***	−0.0521***	0.5301
	(0.0006)	(0.0052)	(0.0052)	(0.0057)	(0.0855)	(0.0205)	(0.0038)	(0.0075)	(0.4051)
TS	−0.1715***	0.0275***	2.1308***	−0.7347***	2.1311***	0.0471***	0.0166	−0.114***	−1.6507***
	(0.0267)	(0.0086)	(0.9901)	(0.0577)	(0.6307)	(0.0026)	(0.0151)	(0.0218)	(0.0743)
REER	0.0377***	−0.0454***	−0.2493***	0.0762***	−0.4482***	0.0355*	0.0184***	−0.0248***	−0.1299
	(0.0073)	(0.0055)	(0.0199)	(0.0236)	(0.1195)	(0.0205)	(0.0040)	(0.0065)	(0.2736)
TCP	0.0352*	−0.0170	−0.0101	0.0012***	−0.0032	0.2355***	0.0031	−0.0026	0.0252***
	(0.0204)	(0.0150)	(0.2303)	(0.0001)	(0.0301)	(0.0205)	(0.0021)	(0.0036)	(0.0041)
GFCF	−0.5976**	−0.8526***	−0.5026***	−0.5004***	−0.7006***	−0.9076***	−1.4016**	2.9906*	0.0136
	(0.2321)	(0.0624)	(0.0039)	(0.0002)	(0.0435)	(0.0302)	(6.1307)	(1.7206)	(0.0464)
Constant	40.2927***	39.6876***	52.1007***	15.0412***	92.1427***	4.2021***	−0.0032***	8.1306***	−3.4511
	(2.5051)	(2.0025)	(19.039)	(2.7516)	(17.3333)	(0.3033)	(0.0031)	(2.9015)	(3.6101)
Obs.	1440	1440	2880	1440	1440	2880	7.529***	20.55***	11.5234***
Wald test	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Note: This table presents robustness checks using three approaches: the Feasible Generalized Least Squares (FGLS) estimator, an alternative measure of financial stability (FS), and Panel-Corrected Standard Errors (PCSE) regression. Columns (1–3) report results using FGLS, with the *Z*-score (FSZS) as the dependent variable. Columns (4–6) use the ratio of M2 to GDP as an alternative measure of FS. Columns (7–9) report PCSE regression estimates, with FSZS as the dependent variable. Standard errors are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

4.6.2 | Spline Regression Estimation of Smooth Nonlinearities Between EPU and FS

As a final robustness check, we use spline regression to capture possible threshold effects and nonlinear dynamics that standard polynomial models may not reflect. This method enables flexible curve fitting and facilitates the detection of turning points in economic relationships (Sangalli et al. 2013). Figures 6–8 present the spline regression results for emerging markets, developed economies, and the full sample. In emerging markets, the spline curve reveals a distinct U-shaped pattern. Financial stability declines as EPU rises, reaches a minimum, and then starts to improve. This suggests that higher levels of uncertainty may lead countries to introduce structural reforms or regulatory changes that support stability. In developed economies, the curve appears flatter, and the U-shape is less evident, indicating stronger resilience due to robust institutions and deep financial systems. In the full sample, the U-shape remains evident, with recovery primarily driven by emerging markets. These results support the hypothesis of nonlinear and asymmetric EPU effects, shaped by institutional quality and development levels.

4.7 | Discussion, Policy Implications, and Limitations

4.7.1 | Discussion of the Study

This study examines the impact of EPU on FS across 12 developed and emerging economies from 2003 to 2022. The analysis reveals a significant nonlinear relationship between EPU and FS. At low to moderate levels, increases in EPU reduce FS, reflecting heightened market volatility, reduced investment, and increased risk aversion. However, beyond a critical threshold, further increases in EPU are associated with improvements in FS, suggesting that firms and regulators respond adaptively to sustained uncertainty by implementing risk mitigation strategies and strengthening financial buffers. Sub-sample analyses show that this nonlinear effect proves more pronounced in emerging economies. Weaker institutional frameworks and greater exposure to external shocks in these markets magnify sensitivity to policy uncertainty. Conversely, developed economies exhibit a less pronounced relationship, consistent with their stronger regulatory environments and more mature financial systems, which enable them to better absorb shocks.

These findings align with and extend prior studies. For example Bordo et al. (2016), Nguyen et al. (2021) and Dang et al. (2021) document that policy uncertainty initially destabilizes financial markets but encourages corrective mechanisms as uncertainty persists. The more substantial nonlinear effects in emerging market booms highlight differential vulnerabilities across economies at varying development levels (Phan et al. 2021). In contrast, Wu et al. (2024) find a consistently adverse effect of policy uncertainty on FS, which suggests that regulatory capacity and structural differences across economies shape this relationship. This study contributes to the literature by identifying threshold effects and demonstrating heterogeneity in the EPU–FS nexus between developed and emerging economies.

Theoretically, these results support both fundamental options theory and uncertainty-driven financial cycle models. According to fundamental options theory, economic agents delay irreversible investments during uncertain periods (McDonald and Siegel 1986), initially undermining financial stability (Bae et al. 2025; Baker et al. 2023). Over time, however, policymakers and institutions implement adaptive responses such as regulatory reforms, monetary interventions, and enhanced risk management that stabilize financial systems (Jurado et al. 2015). Furthermore, monetary policy theory explains how economic uncertainty around central bank actions and interest rates can distort credit cycles and induce financial instability (Friedman 1970). Nevertheless, persistent uncertainty may drive institutional and behavioral shifts that ultimately strengthen resilience, consistent with Pastor and Veronesi (2012) financial cycle framework. The stronger nonlinear responses observed in emerging markets underscore the importance of institutional capacity in moderating the adverse effects of uncertainty. Overall, this study contributes to the literature by demonstrating that the EPU–FS relationship is not only nonlinear but also heterogeneous across countries, shaped by institutional and macroeconomic structures. These findings enrich the understanding of how economies respond to uncertainty and inform policy frameworks aimed at enhancing financial stability in uncertain environments.

5 | Concluding Remarks

5.1 | Conclusion

In this paper, we examined the complex relationship between EPU and FS across 12 developed and emerging economies from 2003 to 2022. We applied a comprehensive set of econometric techniques including OLS, Fixed Effects, GLS, quantile regression, spline regression, and dynamic panel GMM. We uncovered a significant nonlinear association whereby EPU negatively affected FS at low to moderate levels but showed a positive effect beyond a critical threshold. This nonlinear pattern proved markedly stronger in emerging economies, highlighting their heightened vulnerability and sensitivity to policy shocks compared to the more resilient financial systems of developed countries. Our findings substantiate and extend fundamental options theory and uncertainty-driven financial cycle models by illustrating how uncertainty initially delays irreversible investments and destabilizes financial systems but eventually triggers adaptive regulatory and institutional responses that restore stability. This theoretical grounding enriches macro-financial literature by revealing the dynamic and heterogeneous effects of policy uncertainty on financial resilience. The study findings remain robust across multiple estimators and alternative FS measures, ensuring the consistency and reliability of our results. The study contributes to highlighting the importance of nonlinearities and cross-country heterogeneity in assessing the impact of EPU on FS.

5.2 | Policy Implications

The nonlinear dynamics identified in this study carry significant policy implications. Policymakers in emerging markets

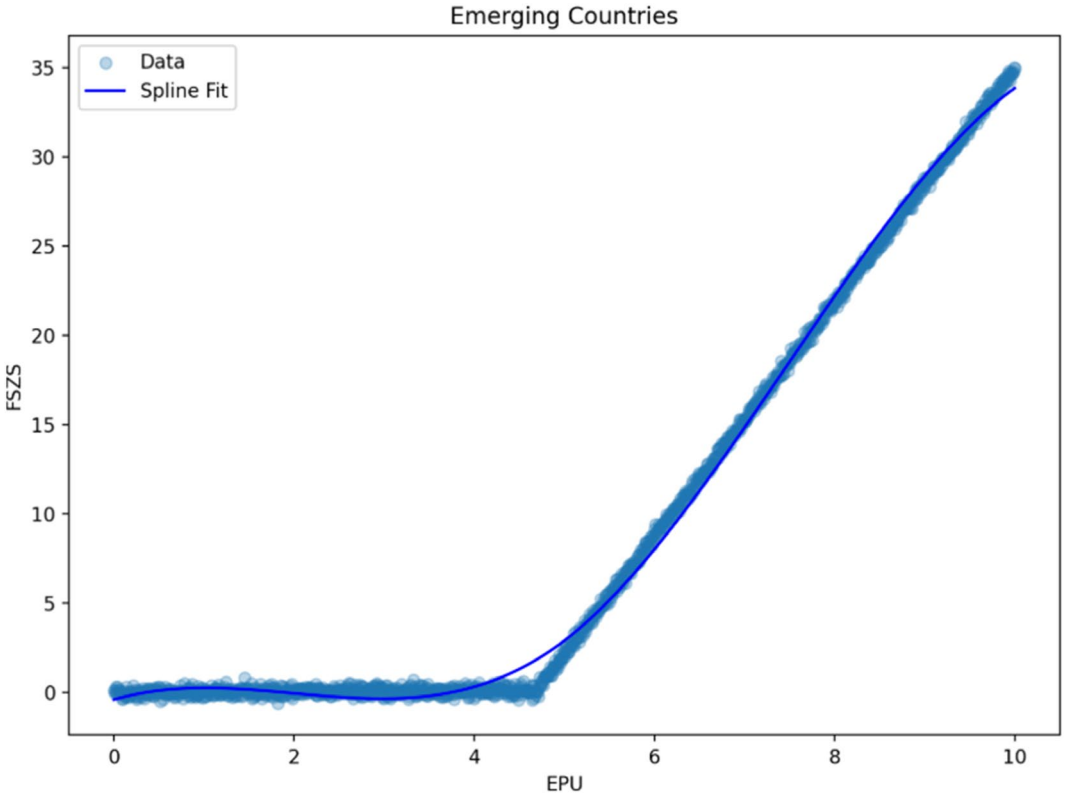


FIGURE 6 | Spline regression plot for emerging countries.

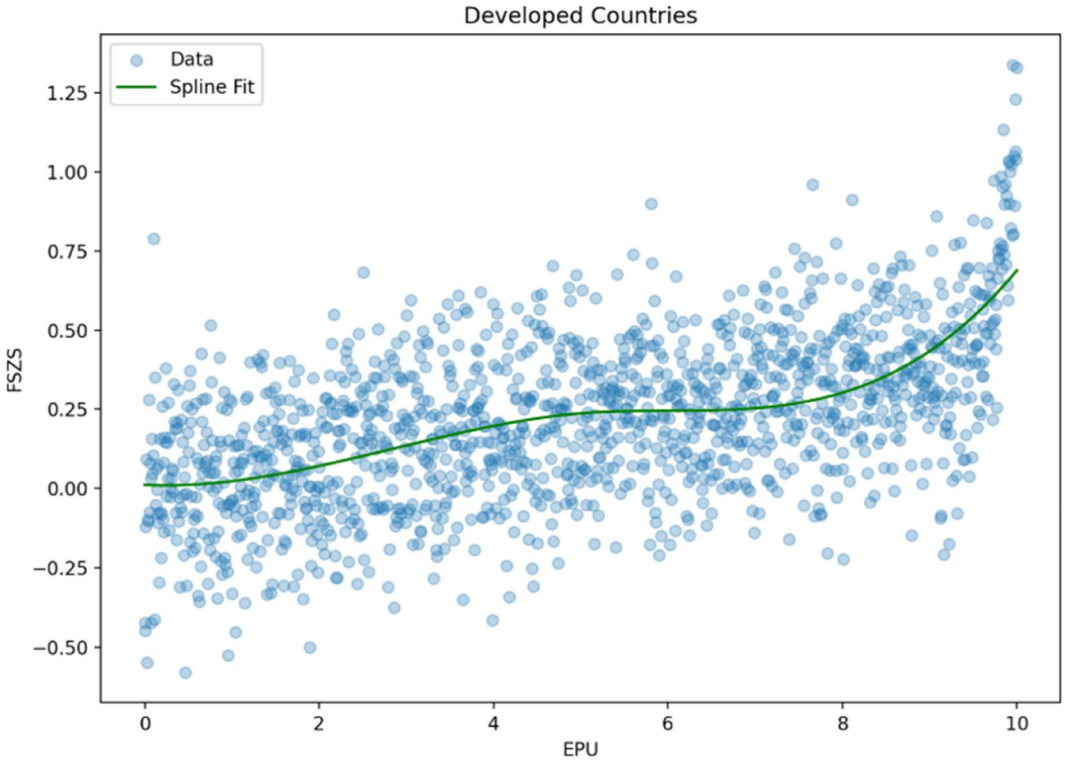


FIGURE 7 | Spline regression plot for developed countries.

should prioritize strengthening institutional quality and regulatory frameworks to enhance resilience against shocks from EPU. Key measures include implementing countercyclical capital buffers, improving liquidity support mechanisms, and enhancing supervisory practices during periods of heightened uncertainty. These steps help mitigate the greater vulnerabilities emerging economies face in the early stages of rising EPU. In developed economies, although financial systems are more robust,

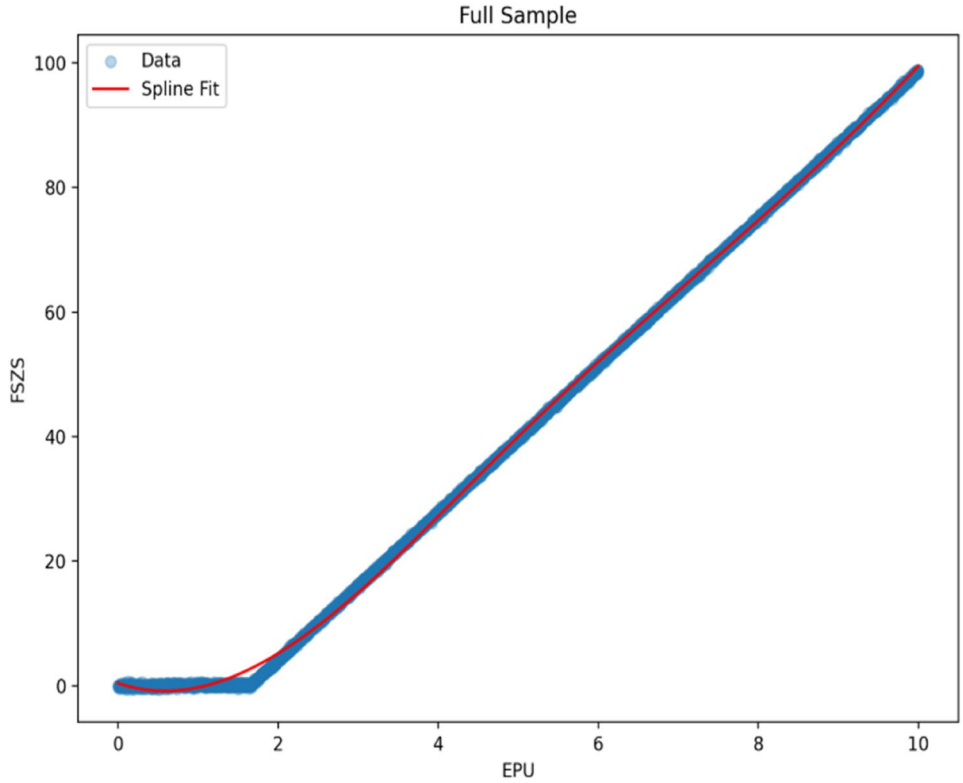


FIGURE 8 | Spline regression plot for full sample.

sustained uncertainty continues to threaten stability. Central banks and regulators must emphasize transparency, clear communication, and forward guidance to reduce market unpredictability. Enhanced disclosure and predictable policy signaling help temper market reactions to uncertainty and lower volatility. Both emerging and developed economies benefit from promoting adaptive investment strategies and financial instruments that enable swift responses to changing policy environments.

Theoretically, this study contributes to the literature by integrating real options theory with empirical evidence of a nonlinear relationship between EPU and FS. It shows that while uncertainty initially destabilizes financial systems, persistent or extreme uncertainty triggers adaptive institutional and policy responses that restore stability. The explicit comparison between developed and emerging economies highlights the moderating role of institutional quality and market maturity in shaping the EPU-FS nexus. This distinction enriches macro-financial stability frameworks by emphasizing heterogeneity across economic contexts and encouraging the development of tailored theoretical models that consider differences in regulatory capacity and vulnerability.

Overall, the findings underscore the importance of tailored macroprudential policies that reflect country-specific vulnerabilities and adaptive capacities. Emerging economies, due to their heightened sensitivity to policy uncertainty, should prioritize strengthening their institutional frameworks and regulatory capacity. Tools such as countercyclical capital buffers, liquidity support, and enhanced supervisory oversight can mitigate the destabilizing effects of uncertainty. Developed economies, despite stronger financial systems, must maintain clear and transparent communication, including forward guidance from central banks, to reduce market volatility during sustained periods of uncertainty. This study advances the understanding of the EPU-FS nexus by providing empirical evidence of a nonlinear relationship and uncovering the varying impacts across different economic contexts.

5.3 | Limitations and Future Directions

This study identifies a nonlinear impact of EPU on FS; however, we acknowledge the following limitations. First, a note of caution is warranted regarding the use of news-based EPU data, which may not fully capture the range of EPU-related exposures that influence FS. Future research should consider alternative or complementary measures of EPU. Second, the macro-level analysis may overlook sector-specific heterogeneity in how industries respond to EPU, which limits the granularity of the findings. Third, the study does not develop explicit economic transmission mechanisms; therefore, future studies should examine the moderating role of digital finance and sustainability practices in mitigating the adverse effects of EPU on FS. Finally, incorporating firm-level EPU exposure, LLM-based ethical policy uncertainty indices, and text-based measures could offer more nuanced insights into the EPU–FS relationship.

Acknowledgments

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Endnotes

[1] World Bank *Z*-score for FS is consistent with existing studies (Barik and Pradhan 2021; Morgan and Pontines 2014; Sethy and Goyari 2022).

[2] M2 includes M1 as well as short-term time deposits in banks and certain money market funds.

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Appendix A

TABLE A1 | Panel Unit Root Test Estimations.

	Emerging countries			Developed countries			Full sample		
At level	**LL&C**	**IPS**	**ADF—Fisher**	**LL&C**	**IPS**	**ADF—Fisher**	**LL&C**	**IPS**	**ADF—Fisher**
FSZS	−2.91***	−1.904**	20.53**	−1.759**	−1.384**	27.21***	−1.85**	−1.384**	27.31***
EPU	−1.568**	−4.159***	51.53***	−2.74***	−4.34***	64.99***	−3.90***	−4.304***	64.909***
EPU^2	−0.370	−6.492***	86.08***	−2.54***	−4.83***	114.31***	−5.90***	−8.803***	104.30***
CPI	3.343	5.434	6.006	−2.91***	−4.07***	81.453***	−2.91***	−4.57***	80.053***
TS	1.050	−0.927	13.53	−0.101	−0.318	12.824	−1.901	−0.018	12.024
REER	−0.613	−0.596	11.91	−0.571	−0.950	15.147	−0.571	−0.050	1.147
TCP	23.318	5.524	2.417	−4.30***	−2.032**	21.061**	−6.37***	−2.002**	21.061**
GFCF	−0.813	−5.001***	113.10***	−1.500*	0.721	6.359	−1.900*	0.701	6.159
1st difference									
CPI	−7.86***	−15.85***	251.75***				−2.91***	−8.07***	81.433***
TS	3.065	−13.65***	200.91***	−22.4***	−21.4***	367.05***	−2.01***	−4.08***	61.203***
REER	−28.7***	−25.98***	468.03***	−31.5***	−27.7***	501.59***	−21.5***	−27.02***	401.09***
GFCF	−4.07***	81.453***	−2.901***	−3.14***	−6.63***	75.15***	−3.02***	−6.503***	75.105***

Note: This table reports panel unit root test results for all variables using three different tests: LL&C: Levin, Lin & Chu test, IPS: Im, Pesaran and Shin test and ADF—Fisher: Augmented Dickey–Fuller Fisher-type test. The tests are applied at both level and first difference to determine the stationarity of the variables. Significance levels: $p < 0.10$ (), $p < 0.05$ (), $p < 0.01$ ().

TABLE A2 | Diagnostic testing.

	Emerging countries			Developed countries			Full sample		
White's test									
H0: Homoskedasticity									
Ha: Unrestricted heteroskedasticity									
chi^2 (43)	321.32			804.89			1402.01		
Prob > chi^2	0.0000			0.0000			0.0000		
Cameron and Trivedi's decomposition									
IM-test									
Source	chi^2	df	*p*	chi^2	df	*p*	chi^2	df	*p*
Heteroskedasticity	321.32	43	0.000	804.89	43	0.000	1019.12	43	0.000
Skewness	126.09	8	0.000	282.8	8	0.000	402.01	8	0.000
Kurtosis	24.17	1	0.000	54.8	1	0.000	204.81	1	0.000
Total	471.59	52	0.000	1142.49	52	0.000	1625.94	52	0.000

	Emerging countries	Developed countries	Full sample
Modified Wald test for group wise heteroskedasticity in fixed effect regression model			
H0: sigma $(i)^2$ = $sigma^2$ for all i			
chi^2 (6)	1693.68	1530.28	2530.11
Prob > chi^2	0.0000	0.0000	0.0000

	Emerging countries	Developed countries	Full sample
Hausman test specification			
Test of H0: Difference in coefficients not systematic			
chi^2 (5) = (b−B)'[(V_b−V_B) ^ (−1)] (b−B)			
chi^2 (6)	283.42	397.36	591.11
Prob > chi^2	0.0000	0.0000	0.0000

	Emerging countries	Developed countries	Full sample
Wooldridge test for autocorrelation in panel data			
H0: no first-order autocorrelation			
F (1,6)	1996.256	474.548	904.011
Prob > F	0.0000	0.0000	0.0000

	Emerging countries		Developed countries		Full sample	
Breusch and Pagan Lagrange multiplier test for random effects						
FSZ [id, *t*] = Xb + u[id] + e [id, *t*]						
Estimated results	Var	SD = sqrt (Var)	Var	SD = sqrt (Var)	Var	SD = sqrt (Var)
FSZ	23.22	4.819	22.706	4.765	29.101	8.061
E	5.497	2.344	9.061	3.01	13.001	9.01
U	0.000	0.000	0.000	0.000	0.000	0.000
Test: Var (u)	0.000		0.000		0.000	
$chibar^2$ (01)	0.000		0.000		0.000	
Prob > $chibar^2$	1.0000		1.0000		1.0000	

Note: Appendix Table A2 gives the results of the diagnostic testing for emerging and developed countries including the sample period 2003-2022.

TABLE A3 | Cross sectional dependence (CD) test.

Variable	Breusch–Pagan LM	Pesaran scaled LM	Bias-corrected scaled LM	Pesaran CD
(1) FSZC	2868.5***	243.9***	243.8***	26.1***
(2) EPU	3090.7***	263.3***	263.2***	48.8***
(3) CPI	9921.7***	857.8***	857.8***	87.8***
(4) TS	3539.1***	302.3***	302.3***	28.4***
(5) REER	3274.7***	279.3***	279.2***	−8.1***
(6) TCP	6956.3***	599.7***	599.7***	61.1***
(7) GFCF	8317.3***	718.2***	718.1***	75.2***

Note: Table 3 provides the results of diagnostic testing for the cross-sectional dependence among all chosen explained and explanatory variables for 78 developing nations from 2009 to 2023, based on a total of 1170 observations. ()***Significant at ***$p < 0.01$.

TABLE A4 | Co integration test.

Cointegration test	Effect of EPU on FS	Non-linearity of EPU2
Westerlund test		
Variance Ratio	11.02*	10.89*
Pedroni test		
Modified Phillips–Perron *t*	−46.21*	−45.63*
Phillips–Perron *t*	−70.34	−69.12
Augmented Dickey–Fuller *t*	−58.47*	57.81*
Kao test		
Modified Dickey–Fuller *t*	−19.11**	−15.07*
Dickey–Fuller *t*	−9.02*	−7.71**
Augmented Dickey–Fuller *t*	−14.03***	−11.91**
Unadjusted Modified Dickey–Fuller *t*	−11.12*	−9.38*
Unadjusted Dickey–Fuller *t*	−16.47*	−13.52*

Note: This table reports the test results, which reject the null hypothesis of no cointegration in both dimensions. The findings confirm the existence of a long-run equilibrium relationship between EPU and FSZS, with nonlinear effects captured by the squared term of EPU (EPU^2). The evidence supports the estimation of a long-run cointegration model.

TABLE A5 | CS-ARDL estimation.

	Linear (EPU → FS)	Nonlinear (EPU & EPU^2 → FS)
	(1)	(2)
Variables	FSZS	FSZS
Long term		
EPU	−0.214***	−0.198***
	(0.058)	(0.061)
EPU2		0.037**
		(0.015)
CPI	−0.087**	−0.091**
	(0.041)	(0.043)
TS	0.126**	0.119**
	(0.054)	(0.053)
REER	−0.061**	−0.056**
	(0.027)	(0.026)
TCP	0.174***	0.056***
	(0.041)	(0.026)
GFCF	0.095***	0.092***
	(0.024)	(0.026)
Short term		
Error Correction Term (ECTt-1)	−0.417***	−0.428***
	(0.067)	(0.083)
ΔEPU	0.097***	−0.89***
	(0.026)	(0.028)
ΔEPU2		0.015**
		(0.007)
ΔCPI	−0.043**	−0.046**
	(0.019)	(0.020)
ΔTS	0.059**	0.057**
	(0.025)	(0.026)
ΔREER	−0.028*	−0.025**
	(0.015)	(0.014)
ΔTCP	0.083***	0.081***
	(0.021)	(0.020)
ΔGFCF	0.045***	0.043***
	(0.012)	(0.013)
Obs.	2880	2880
R-squared	0.6921	0.7015
Cross sectional dependence	0.000	0.000

Note: Estimated using CS-ARDL ($p = 1$, $q = 1$) following Chudik and Pesaran (2015). All models include cross-sectional averages to address dependence—robust standard errors in parentheses. Negative and significant ECT confirm long-run equilibrium adjustment. Δ-prefixed variables indicate first-differenced (short-run) effects. *, **, *** denote significance at 10%, 5%, 1%.